

Sharp Quantitative Faber–Krahn Stability via a Contradiction-Free Route

Plan 1 / Route A: assembled document

Abstract

We give a proof of the sharp quantitative Faber–Krahn inequality

$$\left(\frac{|\Omega|}{|B_1|}\right)^{2/N} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(N) \mathcal{A}(\Omega)^2,$$

with sharp exponent 2 on the Fraenkel asymmetry $\mathcal{A}$ (equivalently 1/2 on the scale-normalised deficit), in which the stability constant $c_{\text{FK}}(N)$ is an explicit composition of named constants. The sequential-selection / compactness contradiction step in the proof of Brasco–De Philippis–Velichkov [1] is replaced by a single-set quantitative selection principle, and every subsequent step is constructive. The proof is organised into seven self-contained building blocks, each delivered as a separate L^AT_EX document:

- `SelectionPrinciple.tex` — single-set quantitative selection;
- `GraphEntry.tex` — BDV asymmetry $\rightarrow$ global $C^{1,\gamma}$ normal graph;
- `SchauderInterpolation.tex` — $C^{1,\gamma} \rightarrow$ small C^{2,γ_0} ;
- `NearlySphericalClosure.tex` — BDV Theorem 3.3 delivers $Q_\alpha \geq c_*(N)$;
- `SaintVenantAssembly.tex` — small + far regimes glue to sharp Saint–Venant stability inside a fixed ball B_R ;
- `BoundedReduction.tex` — removes the R -dependence by reducing the global case to the bounded one via the constructive BDV truncation ([1, §5]);
- `KohlerJobinTransfer.tex` — Saint–Venant $\rightarrow$ Faber–Krahn via Kohler–Jobin.

The introduction sets the result in the literature [2, 1, 3]; this master document gives the chain of labelled theorems and the final assembled statement.

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1 Introduction

1.1 The Faber–Krahn inequality

Let $\Omega \subset \mathbb{R}^N$, $N \geq 2$, be an open set of finite Lebesgue measure and let

$$\lambda_1(\Omega) = \min_{u \in H_0^1(\Omega) \setminus \{0\}} \frac{\int_{\Omega} |\nabla u|^2 dx}{\int_{\Omega} u^2 dx}$$

be the first Dirichlet eigenvalue of the Laplacian on Ω . The *Faber–Krahn inequality* asserts that among open sets of a prescribed volume the ball uniquely minimises λ_1 : writing B^* for the ball with $|B^*| = |\Omega|$,

$$\lambda_1(\Omega) \geq \lambda_1(B^*), \quad (1)$$

with equality if and only if Ω is a ball (up to a set of capacity zero). This was conjectured by Lord Rayleigh in the late nineteenth century in the planar case and proved in general dimension by Faber [6] and Krahn [7] using Schwarz symmetrisation and the Pólya–Szegő principle. Rigidity reflects the fact that the only equality cases of Schwarz symmetrisation in the gradient norm are radially symmetric functions; modern accounts appear in [8, 9, 10].

Once (1) has been proved with rigidity, it is natural to ask for a *stability* statement: if $\lambda_1(\Omega)$ is only slightly larger than $\lambda_1(B^*)$, then Ω should be in a quantitative sense close to a ball. The natural way to measure the deviation from spherical symmetry is the *Fraenkel asymmetry*

$$\mathcal{A}(\Omega) = \inf_{x_0 \in \mathbb{R}^N} \frac{|\Omega \Delta (B^* + x_0)|}{|B^*|} \in [0, 2), \quad (2)$$

which vanishes precisely on translates of the ball. A quantitative Faber–Krahn inequality is then an estimate of the form

$$\left(\frac{|\Omega|}{|B_1|} \right)^{2/N} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c \omega(\mathcal{A}(\Omega)), \quad (3)$$

for some modulus $\omega : [0, 2) \rightarrow [0, \infty)$ vanishing only at 0. The first systematic study of such bounds is due to Hansen and Nadirashvili [11], with parallel contributions by Melas [12]; both gave bounds of the shape (3) only for restricted classes of sets and with a non-sharp modulus. The first stability result valid for every open set in terms of the Fraenkel asymmetry was obtained by Bhattacharya [13] in dimension $N = 2$ (exponent 3). Fusco, Maggi and Pratelli [14], exploiting their sharp quantitative isoperimetric inequality [2], extended this to all dimensions with exponent 4 on $\mathcal{A}$; see also [3, 15].

1.2 The sharp exponent

The optimal modulus in (3) is dictated by the family of linear ellipsoidal perturbations of the ball, $\Omega_{\varepsilon} = \{(x', x_N) \in \mathbb{R}^N : |x'|^2 + (1 + \varepsilon)x_N^2 \leq 1\}$, $0 < \varepsilon \ll 1$, for which a direct computation yields

$$\mathcal{A}(\Omega_{\varepsilon}) \simeq \varepsilon, \quad \lambda_1(\Omega_{\varepsilon}) - \lambda_1(B^*) \simeq \varepsilon^2;$$

see [1, §1.2]. Consequently the sharp exponent is 2 on the asymmetry, equivalently 1/2 on the spectral deficit, and the sharp quantitative Faber–Krahn inequality is the conjecture

$$\left(\frac{|\Omega|}{|B_1|}\right)^{2/N} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c(N) \mathcal{A}(\Omega)^2. \quad (4)$$

The exponent 2 is natural because the second variation of λ_1 around the ball is non-degenerate on the subspace of volume- and barycentre-preserving normal perturbations. The conjecture (4) sits at the spectral end of a family of sharp quantitative problems (isoperimetric, Sobolev, log-Sobolev, Pólya–Szegő, Wulff, Riesz potentials), the prototype being the sharp quantitative isoperimetric inequality of Fusco–Maggi–Pratelli [2], with later proofs and refinements by Cicalese–Leonardi [16], Figalli–Maggi–Pratelli [17], and Fusco–Julin [18]; see [19, 20] for surveys.

1.3 The Brasco–De Philippis–Velichkov theorem

The conjecture (4) was settled in 2015 by Brasco, De Philippis and Velichkov [1]:

Theorem 1.1 (Brasco–De Philippis–Velichkov; [1, Main Theorem]). *For every $N \geq 2$ there is a constant $\sigma = \sigma(N) > 0$ such that*

$$|\Omega|^{2/N} \lambda_1(\Omega) - |B^*|^{2/N} \lambda_1(B^*) \geq \sigma \mathcal{A}(\Omega)^2$$

for every open set $\Omega \subset \mathbb{R}^N$ of finite measure.

The strategy of [1] has two main pillars. First, an application of the Kohler–Jobin rearrangement inequality [21, 3] reduces (4) at fixed volume to the analogous *sharp Saint–Venant stability*

$$E(\Omega) - E(B_1) \geq \sigma \mathcal{A}(\Omega)^2, \quad |\Omega| = |B_1|, \quad (5)$$

where $E(\Omega) = \min\{\frac{1}{2} \int |\nabla u|^2 - \int u : u \in H_0^1(\Omega)\} = -\frac{1}{2}T(\Omega)$. Second, (5) is proved by the *selection principle* introduced by Cicalese and Leonardi [16] for the isoperimetric inequality and refined by Acerbi–Fusco–Morini [22] in the nonlocal isoperimetric setting. Arguing by contradiction, one supposes a sequence Ω_j violates (5) with $\mathcal{A}(\Omega_j) \rightarrow 0$ and constructs an improved sequence U_j by penalising the energy along BDV’s smooth asymmetry α :

$$U_j \in \arg \min \left\{ E(\Omega) + \sqrt{\varepsilon_j^2 + \sigma(\alpha(\Omega) - \varepsilon_j)^2} : |\Omega| = |B_1| \right\}. \quad (6)$$

The Alt–Caffarelli free-boundary regularity machinery [23] applied to the perturbed Bernoulli problem (6) upgrades L^1 convergence $U_j \rightarrow B_1$ to $C^{2,\gamma}$ convergence of ∂U_j as normal graphs. Once U_j is a smooth normal graph over B_1 with small $C^{2,\gamma}$ norm, the explicit second-order Fuglede-type expansion [24, 1] of the Saint–Venant energy — BDV’s nearly spherical Theorem 3.3 — yields the desired contradiction.

1.4 The cost of the contradiction step

The proof of Theorem 1.1 just sketched is conceptually clean but inherently *non-constructive*: the constant $\sigma(N)$ in the main theorem is produced by a compactness–contradiction argument applied to the penalised sequence U_j . Three layers of non-effectiveness compound:

- (a) The selection step (6) works only “for j large” and the threshold is not localised.
- (b) The smooth convergence $\partial U_j \rightarrow \partial B_1$ in $C^{2,\gamma}$ is obtained as a subsequential limit; no explicit threshold is given at which the graph entry holds.
- (c) The contradiction itself uses the limit set U_∞ , not any specific U_j , so even if (a) and (b) were quantified the final constant $\sigma(N)$ would still be inaccessible.

As a consequence, although Theorem 1.1 is fully proved as a mathematical statement, the constant $\sigma(N, R)$ (restricted to sets in a fixed ball B_R) is not extractable from the proof, even in principle.

This phenomenon is well known in the literature on sharp quantitative stability and is not specific to [1]: every proof based on penalisation + selection + Alt–Caffarelli regularity that we are aware of shares the same feature; see the surveys [19, 20].

1.5 Main result and contribution

We give a proof of the sharp quantitative Faber–Krahn inequality in which the contradiction step is replaced by a single-set quantitative selection map and every subsequent step is constructive. Our main theorem is the following.

Theorem 1.2 (Sharp quantitative Faber–Krahn, constructive form). *Let $N \geq 2$. There exists a constant $c_{\text{FK}}(N) > 0$, depending only on the dimension, given explicitly by the closed-form expression (12) below as a composition of the named constants of Theorems 3.1–3.7, such that*

$$\left(\frac{|\Omega|}{|B_1|} \right)^{2/N} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(N) \mathcal{A}(\Omega)^2$$

for every open $\Omega \subset \mathbb{R}^N$ of finite positive measure. The exponent 2 on $\mathcal{A}$ is sharp.

The novelty is not the inequality itself — which is [1] — but its *constant*. Theorem 1.2 provides a *truly quantitative* version of the BDV theorem: $c_{\text{FK}}(N)$ is presented as an explicit composition of N -dependent quantities, each of which is the named constant of a quantitative ingredient (BDV Lemmas 4.2, 4.5, 4.9, 4.12, 4.16, 5.1, 5.2, 5.3, BDV Theorems 3.3 and 4.18; the dimensional constants L_N, T_N ; the Lieberman–Schauder constant; the Gagliardo–Nirenberg interpolation constant). We emphasise that we do not compute $c_{\text{FK}}(N)$ to a number: the BDV black boxes themselves are not numerically evaluated. What we deliver is a contradiction-free chain whose constants could in principle be computed if the BDV constants were.

The conceptual gain is the following. In [1] the BDV penalised functional (6) is run along a sequence of near-extremals to produce a limit set. Here we run it *once*, on a single near-extremal Ω_0 , with the penalisation tuned to Ω_0 itself:

$$J_{\Omega_0}(U) := E(U) + f_{\tilde{\eta}}(|U|) + \sqrt{\varepsilon^2 + \tau^2(\alpha(U) - \varepsilon)^2}, \quad \varepsilon := \alpha(\Omega_0), \quad \tau = q^{1/4},$$

where $q := Q_\alpha(\Omega_0) = (E(\Omega_0) - E(B_1))/\alpha(\Omega_0)$ is the Saint–Venant quotient of Ω_0 . The direct method produces a single minimiser U_* whose volume-normalised companion $\tilde{\Omega}$ retains the same asymmetry order as Ω_0 and whose quotient is at most twice that of Ω_0 up to an (N, R) -explicit factor. The BDV regularity package, applied to this one set, then provides a $C^{1,\gamma}$ normal graph parametrisation of $\partial\tilde{\Omega}$ at an explicit threshold; the Kinderlehrer–Nirenberg hodograph straightening [4] with Lieberman’s oblique Schauder bootstrap [5] promote this to a uniform $C^{5,\gamma}$ bound; Gagliardo–Nirenberg interpolation between $\|g\|_{L^\infty}$ (small) and $\|g\|_{C^{5,\gamma}}$ (large but controlled) reaches the small C^{2,γ_0} regime in which BDV’s nearly spherical Theorem 3.3 applies directly to $\tilde{\Omega}$. No subsequence, no limit, no contradiction.

1.6 Structure of the paper

The argument is organised as a chain of seven self-contained building blocks, each proved in a companion file and quoted as a labelled theorem in Section 3. Schematically,

$$\underbrace{\text{(S)}}_{\text{selection}} \rightarrow \underbrace{\text{(G)}}_{\text{graph entry}} \rightarrow \underbrace{\text{(I)}}_{\text{Schauder + interpolation}} \rightarrow \underbrace{\text{(N)}}_{\text{nearly spherical}}$$

operates in the small-asymmetry regime, producing the linear bound $E(\Omega_0) - E(B_1) \geq q_*(N, R)\alpha(\Omega_0)$ on $\{\alpha \leq \alpha_{\text{small}}\}$. Combined with BDV Lemma 4.2 this yields the small-asymmetry portion of sharp Saint–Venant stability. The far regime is covered by BDV’s rearrangement-based Lemma 5.2 (the suboptimal $\mathcal{A}^4$ bound, which contains no contradiction); the two regimes are glued at an explicit threshold supplied by BDV Lemma 4.2. The SAINTVENANTASSEMBLY block (Theorem 3.5) gives (5) with $c_{\text{SV}}(N, R)$ explicit. The Kohler–Jobin rearrangement transfer (Theorem 3.7), in the constructive form due to Brasco [3], takes Saint–Venant stability to Faber–Krahn stability with a multiplier $4L_N/((N+2)T_N)$ that is universal in N . Section 4 states and proves the assembled Theorem 4.1, and Section 5 summarises what replaces the contradiction.

1.7 Related literature

Sharp quantitative stability of geometric and spectral inequalities is a vast subject. The prototype is the sharp quantitative isoperimetric inequality, first proved by Fusco–Maggi–Pratelli [2] by symmetrisation, then reproved by Cicalese–Leonardi [16] via the selection principle, and by Figalli–Maggi–Pratelli [17] via optimal transport; Fusco–Julin [18] gave a strong form involving an oscillation index. Sharp quantitative Pólya–Szegő, Sobolev and log-Sobolev inequalities were established by Cianchi–Fusco–Maggi–Pratelli [26]. Stability for Wulff shapes is due to Figalli–Maggi–Pratelli and Neumayer [27]. In the spectral direction, Brasco–Pratelli [28] treated the second eigenvalue and the Szegő–Weinberger inequality; recent work of Allen–Kriventsov–Neumayer [25] addresses Faber–Krahn-type inequalities via Alt–Caffarelli–Friedman monotonicity formulas.

Two strands of recent work directly motivate the present paper. First, the use of regularity theory *quantitatively*: the realisation that free-boundary regularity packages underlying selection-principle proofs can often be made effective on a single set if one is content with (N, R) -explicit constants. Second, the technique of Fuglede-style expansion for energy functionals around the ball [24, 22, 1], which converts nearly spherical stability into a coercivity statement for a self-adjoint operator on ∂B_1 . The present paper combines these ideas with the specific BDV proof: the BDV penalised functional is run once, the BDV regularity package is applied as a black box, the BDV nearly spherical theorem is invoked directly on the single selected set, and the resulting Saint–Venant stability is transferred to Faber–Krahn via the Brasco–Kohler–Jobin rearrangement. The outcome is a constructive constant chain for the sharp Faber–Krahn inequality.

Remark 1.3 (Reduction to bounded sets). The chain $(S) \rightarrow (G) \rightarrow (I) \rightarrow (N)$ of Theorems 3.1–3.4 produces the sharp Saint–Venant stability inside a fixed ball B_R with constant $c_{\text{SV}}(N, R)$. The BOUNDEDREDUCTION block (Theorem 3.6) reduces the global case to the bounded case at the dimensional radius $R_*(N) := \max\{d(N), 2\}$, using BDV’s constructive truncation lemma [1, Lemma 5.3]. The constants $d(N), \delta(N), C_9(N)$ of that lemma are themselves obtained by De Giorgi-type iteration and a cut-off competitor, with no contradiction step. As a consequence the constant $c_{\text{SV}}^{\text{glob}}(N)$ (and hence $c_{\text{FK}}(N)$) depends only on N .

2 Notation and standing hypotheses

Fix $N \geq 2$ and $R \geq 2$. We use the standard normalisations:

- $u_\Omega \in H_0^1(\Omega)$ is the torsion potential, $-\Delta u_\Omega = 1$ in Ω , $u_\Omega = 0$ on $\partial\Omega$;
- $T(\Omega) = \int_\Omega u_\Omega \, dx$, the torsional rigidity;

- $E(\Omega) = -T(\Omega)/2$, the BDV-normalised Saint–Venant energy:

$$E(\Omega) = \min_{v \in W_0^{1,2}(\Omega)} \left(\frac{1}{2} \int |\nabla v|^2 - \int v \right);$$

- $\lambda_1(\Omega)$, the first Dirichlet eigenvalue of $-\Delta$;
- $\mathcal{A}(\Omega)$, the Fraenkel asymmetry, $\inf_{x_0} |\Omega \Delta(B^* + x_0)|/|B^*|$, where B^* is the ball with $|B^*| = |\Omega|$; $\mathcal{A} \in [0, 2)$;
- $\alpha(\Omega)$, the BDV smooth asymmetry of [1, Def. 4.1];
- $Q_\alpha(\Omega) = (E(\Omega) - E(B_1))/\alpha(\Omega)$ at fixed volume $|\Omega| = |B_1| = \omega_N$.

$L_N = \lambda_1(B_1)$ and $T_N = T(B_1) = |B_1|/(N(N+2))$ are universal dimensional constants. We write B_R for the open ball of radius R about 0.

3 The chain of building blocks

Each block is stated here in the form it is proved in the corresponding companion file. Constants depend only on (N, R) unless indicated.

3.1 Block I: single-set quantitative selection

Theorem 3.1 (SELECTIONPRINCIPLE; see `SelectionPrinciple.tex`). *There exist $\varepsilon_{\text{sel}}(N, R), q_{\text{sel}}(N, R) > 0$ and $C_{\text{sel}}(N, R) > 0$ such that for every quasi-open $\Omega_0 \subset B_R$ with $|\Omega_0| = |B_1|$, $\varepsilon := \alpha(\Omega_0) \leq \varepsilon_{\text{sel}}$, and $q := Q_\alpha(\Omega_0) \leq q_{\text{sel}}$, the penalised functional $J(U) = F_{\hat{\eta}}(U) + \sqrt{\varepsilon^2 + \tau^2(\alpha(U) - \varepsilon)^2}$ with $\tau = q^{1/4}$ admits a minimiser $U_* \subset B_R$ whose volume-normalised companion $\tilde{\Omega} := r_*^{-1}U_*$, $r_* = (|U_*|/|B_1|)^{1/N}$, satisfies:*

- (i) $|\tilde{\Omega}| = |B_1|$, $\frac{1}{2}\varepsilon \leq \alpha(\tilde{\Omega}) \leq \frac{3}{2}\varepsilon$;
- (ii) $Q_\alpha(\tilde{\Omega}) \leq 2C_{\text{sel}}(N, R) Q_\alpha(\Omega_0)$;
- (iii) $\tilde{\Omega}$ inherits the BDV penalised-minimiser regularity package (density estimates, Lipschitz nondegeneracy, smooth Bernoulli coefficient);
- (iv) the Bernoulli law on $\partial\tilde{\Omega}$ takes the volume-rescaled form $|\nabla \tilde{u}|^2 = \tilde{\Lambda} + \tilde{a}_\sigma \tilde{\Psi}$.

The proof is constructive: existence is by the direct method (BDV Lemma 4.6 used as a quantitative black box), and parts (i)–(iv) are explicit estimates from the BDV regularity package and the volume scaling $E(\mu V) = \mu^{N+2}E(V)$.

3.2 Block II: graph entry

Theorem 3.2 (GRAPHENTRY; see `GraphEntry.tex`). *There exist $\alpha_{\text{graph}}(N, R)$, $M_1(N, R)$, $C_H''(N, R) > 0$ and $\gamma = \gamma(N, R) \in (0, 1)$ such that every selected $\tilde{\Omega} \subset B_R$ (Theorem 3.1) with $\alpha(\tilde{\Omega}) \leq \alpha_{\text{graph}}$ is, after translation by its Fraenkel-optimal centre, a global $C^{1,\gamma}$ normal graph over ∂B_1 :*

$$\partial\tilde{\Omega} = \{(1 + g(\theta))\theta : \theta \in \partial B_1\},$$

with $\|g\|_{C^{1,\gamma}(\partial B_1)} \leq M_1(N, R)$ and $\|g\|_{L^\infty(\partial B_1)} \leq C_H''(N, R) \alpha(\tilde{\Omega})^{1/(2N)}$. The chain is asymmetry $\rightarrow$ Hausdorff (BDV Lemma 4.2 + density) $\rightarrow$ Alt–Caffarelli flatness (BDV Theorem 4.18) $\rightarrow$ tubular patching. The threshold $\alpha_{\text{graph}}(N, R) = \min\{\alpha_0, (h_F/C_H')^{2N}, (h_F/C_{\text{Fr}}')^2\}$ is explicit.

3.3 Block III: Schauder bootstrap and interpolation

Theorem 3.3 (SCHAUDERINTERPOLATION; see `SchauderInterpolation.tex`). *Fix $0 < \gamma_0 \leq 1$. There exist $M_5(N, R)$ and an explicit threshold $\alpha_{\text{sph}}(N, R, \gamma_0)$ (closed form in upstream constants) such that the graph g of Theorem 3.2 satisfies*

$$\alpha(\tilde{\Omega}) \leq \alpha_{\text{sph}}(N, R, \gamma_0) \implies \|g\|_{C^{2,\gamma_0}(\partial B_1)} \leq \delta_{\text{sph}}(N, \gamma_0),$$

where $\delta_{\text{sph}}(N, \gamma_0)$ is the smallness threshold of BDV Theorem 3.3. The method is the Kinderlehrer–Nirenberg hodograph straightening of $\tilde{u}$ on $\tilde{\Omega}$, the oblique Schauder bootstrap of Lieberman to $\|g\|_{C^{5,\gamma}} \leq M_5(N, R)$, and the Hölder interpolation $\|g\|_{C^{2,\gamma_0}} \leq C_{\text{GN}} \|g\|_{L^\infty}^{1-\theta} \|g\|_{C^{5,\gamma}}^\theta$ with $\theta = (2 + \gamma_0)/(5 + \gamma) \leq 1/2$.

3.4 Block IV: nearly spherical closure

Theorem 3.4 (NEARLYSPHERICALCLOSURE; see `NearlySphericalClosure.tex`). *There is $c_*(N) > 0$, given explicitly by $c_*(N) = 1/(32 N^2 C_3(N))$ where $C_3(N)$ is the BDV Lemma 4.2(iii) constant, such that every nearly spherical $\tilde{\Omega} = (1 + g)B_1$ with*

$$\|g\|_{C^{2,\gamma_0}} \leq \delta_{\text{sph}}(N, \gamma_0), \quad |\tilde{\Omega}| = |B_1|, \quad x_{\tilde{\Omega}} = 0,$$

satisfies $Q_\alpha(\tilde{\Omega}) \geq c_*(N)$, equivalently $E(\tilde{\Omega}) - E(B_1) \geq c_*(N)\alpha(\tilde{\Omega})$. Volume / barycenter perturbations are absorbed by an $O(\|g\|_{L^2})$ correction at the cost of a fixed dimensional factor. The proof is a direct substitution into BDV Theorem 3.3 ($E(\Omega) - E(B_1) \geq (1/32N^2)\|g\|_{H^{1/2}}^2$) followed by the trace-Poincaré comparison and BDV Lemma 4.2(iii) ($\alpha \leq C_3\|g\|_{L^2}^2$).

3.5 Block V: assembly to sharp Saint–Venant stability

Composing Blocks I–IV gives the linear-in- α inequality $E(\Omega_0) - E(B_1) \geq q_*(N, R)\alpha(\Omega_0)$ on a small- α regime, with $q_*(N, R) = c_*(N)/(2C_{\text{sel}}(N, R))$. Combining with BDV Lemma 4.2(i) ($\mathcal{A}^2 \leq (C_1/\omega_N^2)\alpha$) gives the quadratic form $E - E(B_1) \geq \kappa_{\text{near}}(N, R)\mathcal{A}^2$. The far regime is covered by BDV Lemma 5.2 ($E - E(B_1) \geq (C'_8)^{-1}\mathcal{A}^4$), and the two regimes glue at a threshold $a_{\text{near}} = a_{\text{far}}$ supplied by BDV Lemma 4.2(ii).

Theorem 3.5 (SAINTVENANTASSEMBLY; see `SaintVenantAssembly.tex`). *There exists $c_{\text{SV}}(N, R) > 0$, given by (8) below, such that for every open $\Omega \subset B_R$ with $|\Omega| = |B_1|$,*

$$E(\Omega) - E(B_1) \geq c_{\text{SV}}(N, R) \mathcal{A}(\Omega)^2. \quad (7)$$

Concretely, with $\alpha_{\text{small}}(N, R) := \frac{1}{2} \min\{\varepsilon_{\text{sel}}, \alpha_{\text{graph}}, \alpha_{\text{sph}}\}$ and $a_{\text{far}}(N, R) := \alpha_{\text{small}}(N, R)/(C_2(N, R)\omega_N)$,

$$c_{\text{SV}}(N, R) := \min\left\{ \frac{q_*(N, R) \omega_N^2}{C_1(N)}, \frac{a_{\text{far}}(N, R)^2}{C'_8(N)} \right\}. \quad (8)$$

The exponent 2 on $\mathcal{A}$ is sharp (ellipsoid example).

3.6 Block VI: reduction to bounded sets

The Saint–Venant assembly of Theorem 3.5 delivers a constant $c_{\text{SV}}(N, R)$ that depends on the confining radius R . We remove this dependence by invoking BDV’s constructive truncation lemma [1, Lemma 5.3], which sits inside a chain of three quantitative ingredients of [1, §5] (all constructive: De Giorgi-type L^∞ iteration, rearrangement, and a cut-off competitor; no contradiction). Applied at the dimensional radius $R_*(N) := \max\{d(N), 2\}$ where $d(N)$ is the diameter bound of BDV Lemma 5.3, the truncation step converts the bounded-domain stability into a global one.

Theorem 3.6 (BOUNDEDREDUCTION; see `BoundedReduction.tex`). With $R_*(N) := \max\{d(N), 2\}$ and

$$c_{\text{SV}}^{\text{glob}}(N) := \min\left\{c_{\text{near}}^{\text{glob}}(N), \omega_N^{(N+2)/N} \delta(N)/16\right\} > 0, \quad (9)$$

where $c_{\text{near}}^{\text{glob}}(N) = \omega_N^{(N+2)/N} / [4C_9(N)(1/\tilde{c}_{\text{SV}}(N) + C_9(N))]$ and $\tilde{c}_{\text{SV}}(N) = \omega_N^{-(N+2)/N} c_{\text{SV}}(N, R_*(N))$, for every open $\Omega \subset \mathbb{R}^N$ with $|\Omega| = |B_1|$,

$$E(\Omega) - E(B_1) \geq c_{\text{SV}}^{\text{glob}}(N) \mathcal{A}(\Omega)^2. \quad (10)$$

$c_{\text{SV}}^{\text{glob}}(N)$ depends only on N ; the R -dependence inherited from Theorem 3.5 is internalised at the dimensional radius $R_*(N)$.

3.7 Block VII: Kohler–Jobin transfer

Theorem 3.7 (KOHLEBJOBINTTRANSFER; see `KohlerJobinTransfer.tex`). Given the global sharp Saint–Venant inequality (10), for every open $\Omega \subset \mathbb{R}^N$ of finite positive measure,

$$\left(\frac{|\Omega|}{|B_1|}\right)^{2/N} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(N) \mathcal{A}(\Omega)^2, \quad (11)$$

where

$$c_{\text{FK}}(N) := \min\left\{\frac{4L_N c_{\text{SV}}^{\text{glob}}(N)}{(N+2)T_N}, \frac{L_N(2^{2/(N+2)} - 1)}{4}\right\}. \quad (12)$$

The proof uses the Kohler–Jobin inequality $T(\Omega)\lambda_1(\Omega)^{(N+2)/2} \geq T(B^*)\lambda_1(B^*)^{(N+2)/2}$ (cited from [3, Thm 3.3]), the pointwise transfer $\lambda_1(\Omega) \geq \lambda_1(B^*)(T(B^*)/T(\Omega))^{2/(N+2)}$, the elementary inequality $(1-s)^{-p} - 1 \geq ps$ for $p \in (0, 1], s \in [0, 1]$ (no Taylor), and a two-case split on $T_N - T(\Omega)$ versus $T_N/2$. Note that the input (10) is now unrestricted in Ω , so the output (11) is too.

4 The assembled theorem

Composing Theorems 3.1–3.7:

Theorem 4.1 (Sharp quantitative Faber–Krahn stability, Plan 1 / Route A). Let $N \geq 2$. There exists a constant $c_{\text{FK}}(N) > 0$, depending only on the dimension, given explicitly by (12) as a composition of the named constants of Theorems 3.1–3.7, such that for every open set $\Omega \subset \mathbb{R}^N$ of finite positive measure (where B^* is the ball of the same volume),

$$\boxed{\left(\frac{|\Omega|}{|B_1|}\right)^{2/N} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(N) \mathcal{A}(\Omega)^2.}$$

Equivalently, $\mathcal{A}(\Omega) \leq C(N)((|\Omega|/|B_1|)^{2/N}(\lambda_1(\Omega) - \lambda_1(B^*)))^{1/2}$, and the exponent 1/2 on the deficit is sharp (saturated by volume-preserving ellipsoidal perturbations of B^*).

Proof. Combine Theorem 3.5 (sharp Saint–Venant stability in $B_{R_*(N)}$) with Theorem 3.6 (BOUNDEDREDUCTION removing the R -dependence) and Theorem 3.7 (Kohler–Jobin transfer). Theorem 3.5 is itself the result of composing Theorems 3.1–3.4 at the dimensional radius $R = R_*(N)$ (the chain (S)→(G)→(I)→(N) inside the small- α regime, combined with BDV Lemma 5.2 in the far regime, glued via the two directions of BDV Lemma 4.2). $\square$

5 What replaces the contradiction step

BDV's original proof [1] establishes Theorem 4.1 via a sequential selection plus compactness contradiction: assume a sequence Ω_n violating the inequality with $\mathcal{A}(\Omega_n) \rightarrow 0$, run the BDV penalised functional to select $\tilde{\Omega}_n$, extract an L^1 -convergent subsequence, and reach a contradiction with the nearly spherical theorem at the limit. The compactness step makes the final constant c_{FK} non-extractable.

Route A replaces this with a *single-set* version of the same selection (Theorem 3.1), augmented by the quantitative graph entry, Schauder bootstrap, and interpolation steps (Theorems 3.2–3.3), so that the BDV nearly spherical theorem (Theorem 3.4) can be applied to a single set $\tilde{\Omega}$ rather than to a limit. The Saint–Venant assembly (Theorem 3.5) uses BDV's own Lemma 5.2 (proved by rearrangement, not contradiction) for the far regime. The BOUNDEDREDUCTION step (Theorem 3.6) uses BDV's constructive truncation lemma [1, Lemma 5.3] to remove the R -dependence (no contradiction there either). The Kohler–Jobin transfer (Theorem 3.7) is unchanged from BDV.

Every constant in the chain depends only on N and on the named constants of BDV's quantitative ingredients (Lemmas 4.2, 4.5, 4.9, 4.12, 4.16, 5.1, 5.2, 5.3, Theorems 3.3, 4.18); none is set by a compactness or contradiction argument.

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